

Set B

Question 1:

(a) $Y = \overline{\overline{A} + B} = A\overline{B}$

(b)

$$0.3(5 - V_{\overline{A}} - 1.4)^2 = 0.6[2(5 - 0.6)V_{\overline{A}} - V_{\overline{A}}^2] \Rightarrow V_{\overline{A}} = 0.56 \text{ V}$$

$$\frac{5 - V_Y}{25} = 0.7[2(5 - 0.4)V_Y - V_Y^2] \Rightarrow V_Y = 0.03 \text{ V}$$

$$P_{(1,1)} = (5 - 0) \times \left(0.3(5 - 0.56 - 1.4)^2 + \frac{5 - 0.03}{25} \right) = 14.9 \text{ mW}$$

(c) $V_{\overline{A}} = V_{GS_L} - V_{TN_L} = 5 - 1.4 = 3.6$

$$0.3(5 - V_Y - 1.4)^2 = 0.6[2(3.6 - 0.6)V_Y - V_Y^2] + 0.7[2(5 - 0.4)V_Y - V_Y^2] \Rightarrow V_Y = 0.33 \text{ V}$$

(d) Circuits will vary based on approach.

Question 2:

(a) V_{IH} :

$$\frac{\frac{12 - 0.2}{2}}{\frac{V_{IH} - 0.8}{18} - \frac{0.8 - 0}{95}} = 30 \Rightarrow V_{IH} = 4.49 \text{ V}$$

V_{IL} :

$$\frac{V_{IL} - 0.5}{18} = \frac{0.5}{95} \Rightarrow V_{IL} = 0.59 \text{ V}$$

$$\therefore V_{NH} = 10 - 4.49 = 5.51 \text{ V}$$

$$\therefore V_{NL} = 0.59 - 0.2 = 0.39 \text{ V}$$

$$\therefore V_N = \min(V_{NH}, V_{NL}) = 0.39$$

(b) When driver output high:

$$N_H = \text{floor} \left(\frac{\frac{12 - 10}{2.2}}{\frac{10 - 0.8}{18}} \right) = 1$$

When driver output low:

$$N_L = \text{floor} \left(\frac{\frac{12 - 0.2}{\frac{2.2}{0.2 - 0}}}{18 + 95} \right) = 3,064$$

$$\therefore \text{Overall maximum fanout} = \min(N_H, N_L) = 1$$

(c) Case (0,1), (1,0) & (0,0):

$$P_1 = (7 - 0.2) \times \frac{7 - (0.7 + 0.2)}{5} = 8.3 \text{ mW}$$

Case (1,1):

$$P_2 = (7 - 0) \times \left[\frac{7 - 2.2}{5} + \frac{7 - 0.2}{2.5} \right] = 25.76 \text{ mW}$$

$$\therefore P_{avg} = \frac{3 \times P_1 + P_2}{4} = 12.66 \text{ mW}$$

(d) When driver output high, $N_H = \infty$

When driver output low,

$$N_L = \text{floor} \left(\frac{\left(\frac{7 - 2.2}{5} - \frac{0.8 - 0}{5} \right) \times 30 - \frac{7 - 0.2}{2.5}}{\frac{7 - (0.7 + 0.2)}{5}} \right) = 17$$

$$\therefore \text{Overall maximum fanout} = \min(N_H, N_L) = 17$$

Question 3:

(a) KCL at 'Y',

$$\frac{5 - V_Y}{90} = \frac{V_Y - 0.7}{0.95} \Rightarrow V_Y = 0.74 \text{ V}$$

(b)

$$\beta_R = \frac{\left(6.81 - \frac{9 - 2.3}{5} - \frac{9 - 1}{2.5} - \frac{9 - 0.2}{5} + \frac{0.8 - 0}{5} \right)}{2 \times \frac{9 - 2.3}{5}} = 0.25$$

Question 4:

- (a) $Y = A + B + C$
- (b) 0.2
- (c) Cut-off
- (d) Triode
- (e) OR gate
- (f) NOR (3 input) then NOT
- (g) Triode
- (h) $F = \bar{A}$. So, simply use an NMOS inverter.